

Intelligence-enabled IoT with Gen AI, Powered by Model Compression and Optimization

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Abstract

The convergence of Artificial Intelligence (AI) and the Internet of Things (IoT) is one of the main driving forces of the modern era of smart, autonomous systems. However, deploying AI models on resource-constrained IoT devices poses significant challenges in computation, memory, and power consumption. Model compression and sparsity have been a highly effective approach in the field of optimizing models while maintaining performance. But compared to traditional model compression techniques, its progress in Gen AI and IoT appears relatively slower. To address this research gap and provide insights into ways to promote the integration of model compression and sparsity of Gen AI models in IoT, we explore practical strategies including quantization, pruning, knowledge distillation, low rank factorization (LoRA, QLoRA), KV caching, etc. We aim to provide a road map that traces the evolution of model compression and sparsity from traditional small-scale models to the realm of Gen AI for efficient edge deployment across real-world IoT applications that includes the focus on the industrial applications and the research challenges for deploying intelligence in the IoT eco-system, particularly to enable edge intelligence to take care of the constraints of latency, privacy, and reliability.

CCS Concepts

• **Computer systems organization** → **Embedded systems; Embedded systems.**

Keywords

IoT, Gen AI, LLM, Model compression, Quantization, Pruning, Lottery ticket hypothesis, LoRA, QLoRA, Small Language Model

ACM Reference Format:

Arijit Ukil and Antonio J. Jara. 2025. Intelligence-enabled IoT with Gen AI, Powered by Model Compression and Optimization. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3750571>

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UbiComp Companion '25, Espoo, Finland

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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3750571>

1 Introduction

The integration of AI into resource-constrained edge devices is non-trivial due to the computational and memory requirements of modern deep learning models. As ubiquitous computing evolves towards pervasive intelligence, the ability to embed sophisticated AI models into everyday objects, surfaces, and wearables will define the next frontier of human-centered technology [13, 18]. We position model compression and optimization not merely as engineering challenges, but as the enablers of a future where real-time, personalized, and context-aware AI operates invisibly within the physical world using different tools to design efficient models for heterogeneous, resource-constrained environments through secured IoT-based ecosystem [11].

As we continue to integrate AI into ubiquitous systems, industries are increasingly seeking ways to deploy powerful, real-time AI models at the edge [17]. We understand a critical need in industrial research: enabling scalable, energy-efficient, and responsive AI solutions for resource-constrained devices, catering the need of smart healthcare, smart cities, autonomous vehicles, and industrial automation for implementing sophisticated AI systems in edge devices that often operate in real-time while consuming minimal energy with lesser computing budget [14–16]. We evaluate the cost of high parameterization of the state-of-the-art AI models like LLMs and evaluate different model optimization algorithms that include Lottery Ticket Hypothesis [4, 19], Junk DNA Hypothesis [23], Knowledge Distillation [6], and also Gen AI model-specific pruning algorithms like Wanda [12]. We delve into the optimized model construction techniques including sparse representations that leverages structured and unstructured sparsity to minimize parameter redundancy, Parameter-Efficient Fine-Tuning like LoRA (Low-Rank Adaptation) [7], QLoRA [3] adapters to accomplish efficient fine-tuning for specific tasks. Further, we discuss the inference-time efficiency in transformer architecture through the Key-Value (KV) caching approach [2].

2 Solution Sketch

The importance and the natural significance of AI in the IoT eco-system is well-recognized. We need to comprehend how different IoT applications leverage AI to solve the grand challenges of industrial automation, smart health, smart city, etc are noteworthy [9]. Broadly speaking there are four approaches to deep neural network compression and acceleration [1], quantization, pruning, low rank factorization, and knowledge distillation [15].

2.1 Model Quantization

Model quantization reduces the precision of neural network parameters and activations to lower-bit formats (e.g., INT8, INT4, BF16)

to optimize memory and computational efficiency while preserving accuracy. BF16 (bfloat16) maintains the dynamic range of FP32 with reduced precision, enabling faster training and inference on compatible hardware like TPUs and GPUs¹.

2.1.1 Uniform Quantization. Uniform quantization maps weights to discrete levels. The quantized weight $\mathbf{W}_{\text{quant}}$ is given by:

$$\mathbf{W}_{\text{quant}} = \text{round}\left(\frac{\mathbf{W} - \mathbf{W}_{\min}}{\Delta}\right) \Delta + \mathbf{W}_{\min} \quad (1)$$

where Δ is the quantization step size.

2.1.2 Non-Uniform Quantization. Non-uniform quantization uses variable step sizes. The quantized weight $\mathbf{W}_{\text{quant}}$ is given by:

$$\mathbf{W}_{\text{quant}} = Q(\mathbf{W}) \quad (2)$$

where Q is a non-linear quantization function.

2.1.3 Quantization-Aware Training (QAT). In QAT, quantization process is performed at the time of training the model with the training objective:

$$\min_{\mathbf{W}} \mathcal{L}(\mathbf{W}_{\text{quant}}) \quad (3)$$

2.1.4 Post-Training Quantization (PTQ). PTQ applies quantization after training. The quantized model $\mathbf{W}_{\text{quant}}$ is obtained by:

$$\mathbf{W}_{\text{quant}} = Q(\mathbf{W}_{\text{trained}}) \quad (4)$$

2.1.5 Quantization Error Analysis. Quantization error is analyzed using mean squared error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\mathbf{W}_i - \mathbf{W}_{\text{quant},i})^2 \quad (5)$$

2.2 Model Pruning

In model pruning, the redundant parameters are removed to compress the deep neural network from overparametrization to optimal parametrization. There are different pruning strategies involved: local vs global pruning; structured vs unstructured pruning. Recently, lottery ticket hypothesis is proposed to effectively prune a large deep neural network model [4, 19].

2.2.1 Structured and Unstructured Pruning. Structured pruning removes entire neurons, filters, or channels, while unstructured pruning removes individual weights. The objective for structured pruning can be formulated as:

$$\min_{\mathbf{W}} \|\mathbf{W}\|_0 \quad \text{subject to} \quad \mathcal{L}(\mathbf{W}) \leq \epsilon \quad (6)$$

where $\mathcal{L}$ is the loss function and ϵ is a tolerance level.

2.2.2 Global and Local Pruning. Global pruning applies a threshold across the entire network, while local pruning applies thresholds within specific layers. The global pruning objective can be written as:

$$\min_{\mathbf{W}} \sum_{i,j} \mathbf{1}_{|\mathbf{W}_{ij}| < \tau} \quad (7)$$

where τ is the pruning threshold.

¹https://docs.nvidia.com/cuda/archive/12.2.0/cuda-math-api/group__CUDA__MATH__INTRINSIC__BFloat16.html

2.2.3 Lottery Ticket Hypothesis. The Lottery Ticket Hypothesis suggests that dense neural networks contain subnetworks (winning tickets) such that when trained in isolation it can achieve comparable performance. Formally, let $\mathbf{W}$ be the weights of the dense network and $\mathbf{m}$ be a binary mask. The pruned network is represented as:

$$\mathbf{W}_{\text{pruned}} = \mathbf{m} \odot \mathbf{W} \quad (8)$$

2.3 Low-rank factorization

A matrix or tensor factorization finds a lower dimensional manifold by expressing a large number of learnt filters as a linear combination of a small number of basis filters, which it is based on the equivariant group theory which states that applying a transformation on the convolved version of the input data yields the same result as first applying the transformation on the data and then convolving it.

2.3.1 LoRA. LoRA (Low-Rank Adaptation) factorizes weight matrices into low-rank components. The factorized weight $\mathbf{W}_{\text{LoRA}}$ is given by:

$$\mathbf{W}_{\text{LoRA}} = \mathbf{A}\mathbf{B} \quad (9)$$

where $\mathbf{A}$ and $\mathbf{B}$ are low-rank matrices.

2.3.2 QLoRA. QLoRA (Quantized Low-Rank Adaptation) combines quantization with low-rank factorization. The quantized factorized weight $\mathbf{W}_{\text{QLoRA}}$ is given by:

$$\mathbf{W}_{\text{QLoRA}} = Q(\mathbf{A})Q(\mathbf{B}) \quad (10)$$

2.4 Knowledge Distillation

The fourth approach is called knowledge distillation [6], which is often explained through a teacher-student network, where, the student is a smaller network that learns to approximate the teacher. In fact, knowledge distillation seems to be one of the most promising approaches in AI-IoT application development [5], which transfers the knowledge of a teacher model to a relatively smaller model, called the student model by minimizing the distillation loss. The distillation loss is given by:

$$\mathcal{L}_{\text{distill}} = \alpha \mathcal{L}_{\text{student}} + \beta \mathcal{L}_{\text{teacher}} \quad (11)$$

where α and β are weighting factors. Further, the process involves transferring knowledge from the teacher to the student using soft targets and temperature scaling. Soft targets refer to the probability distributions output by the teacher model, offering richer information that attempts to capture the inter-class relationships as well as the model confidence. The soft targets are defined as:

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (12)$$

where z_i is the logit for class i , and T is the temperature parameter. Temperature scaling is used to soften the probabilities produced by the teacher model. Higher temperatures produce softer probabilities. The temperature-scaled probabilities are used as targets for the student model.

2.5 Approaches of model optimization in Gen AI models

Model optimization in Gen AI employs advanced techniques to reduce computational overhead while preserving model performance. Key-Value (KV) caching accelerates autoregressive decoding by storing and reusing attention key and value tensors from prior timesteps, eliminating redundant computation in transformer blocks. LoRA (Low-Rank Adaptation) introduces trainable low-rank decomposition matrices into pre-trained weight layers, enabling efficient fine-tuning with minimal parameter updates [7]. QLoRA extends this by quantizing the base model (e.g., 4-bit precision via NF4) while maintaining LoRA adapters in higher precision, achieving state-of-the-art results with low memory usage [3]. Junk DNA hypothesis assumes a significant fraction of the weights contribute negligibly and task-specific fine-tuning is to be carefully controlled [23]. The Wanda (Weight and Activation-Driven) pruning method selects weights for removal based on their combined magnitude and activation-driven importance, preserving salient information flow [12]. Dedicated foundation model is getting evolved like an edge-cloud collaborative system featuring open-set recognition capabilities with targeted knowledge querying and adaptive edge model customization [22].

2.5.1 KV Caching During Autoregressive Decoding. Caching mechanism avoids recomputation of K and V for previous tokens, significantly improving inference efficiency. At each time step t , we compute: $q_t = x_t W^Q, k_t = x_t W^K, v_t = x_t W^V$ and subsequently update the cache:

$$\begin{aligned} K_{\text{cache}} &\leftarrow [K_{\text{cache}}, k_t] \\ V_{\text{cache}} &\leftarrow [V_{\text{cache}}, v_t] \end{aligned}$$

Then compute attention using cached keys and values:

$$\text{Attention}(q_t, K_{\text{cache}}, V_{\text{cache}}) = \text{softmax}\left(\frac{q_t K_{\text{cache}}^\top}{\sqrt{d_k}}\right) V_{\text{cache}}$$

3 Gen AI at Edge- Research Direction

The deployment of LLMs at the edge represents a transformative shift in the field of Gen AI, enabling intelligent, context-aware, and privacy-preserving applications across a wide range of domains—from mobile devices and wearables to autonomous systems and industrial IoT. **Latency:** In domains such as augmented reality, autonomous navigation, and voice-based interfaces, even a few hundred milliseconds of delay can degrade usability or compromise safety. Techniques such as early exit strategies, adaptive computation, and token pruning offer promising avenues for reducing inference time by dynamically adjusting the computational path based on input complexity.

Power Efficiency: LLMs typically require billions of operations per inference, which translates to significant energy draw. Techniques like SmoothQuant [21] and mixed-precision inference have demonstrated that aggressive quantization can be achieved with minimal accuracy degradation.

Memory Constraints: Knowledge distillation, as exemplified by models like DistilBERT [10] and TinyBERT [8], allows for the

transfer of knowledge from large teacher models to smaller student models. Additionally, sparse attention mechanisms (e.g., Longformer [24], Linformer [20]) reduce memory usage by avoiding full self-attention over long sequences.

- (1) **Unified Learning:** A single FM can process multimodal sensor input and perform classification, prediction, summarization, and control.
- (2) **Semantic Understanding:** Embedded world knowledge enables better context-aware inference for ambiguous or sparse data.
- (3) **Language-based Control:** Natural language commands can be used to configure or program devices.
- (4) **Transferable Intelligence:** Once trained, FMs can generalize across applications and devices.

3.1 Generative AI Applications in IoT

- (1) **Sensor Data Imputation:** Diffusion and GAN-based generative models reconstruct missing signals.
- (2) **Forecasting:** Transformer-based models like FEDformer and TimesNet provide long-term predictions.
- (3) **Simulation:** Generative AI simulates rare edge-case failures for industrial predictive maintenance.
- (4) **Conversational Agents:** Whisper + TinyLLM enables voice control in smart homes and wearables.

3.2 Emerging Research Directions

- (1) **Multimodal IoT LLMs:** Design of small-scale FMs that can reason over diverse sensor types.
- (2) **Edge-friendly Pretraining:** Using smaller, domain-specific corpora (TinyStories, IoT-Bench) to train compressed models.
- (3) **Neurosymbolic Fusion:** Combining symbolic control logic with LLM generalization.
- (4) **On-device Memory:** Lightweight vector DBs and RAG modules embedded on-device for persistent memory.

3.3 Open Research Challenges

- (1) **Tokenization of Time Series:** Need for standardized tokenizers for non-text modalities.
- (2) **Hallucination and Interpretability:** Ensuring LLM outputs are grounded in sensor data.
- (3) **Dynamic Adaptation:** Efficient on-device continual learning remains unsolved.
- (4) **Energy-Aware LLMs:** Sustainable inference under strict power budgets.

4 Open Research Challenges and Future Directions

- (1) **Adaptive Runtime Models-** Models that adapt dynamically to hardware availability, battery status, or context-aware inference requirements can enhance robustness. For example, multi-exit architectures or early-exit networks dynamically terminate inference based on confidence thresholds.

- (2) **Cross-layer Optimization**- Optimization across hardware, compiler, and model simultaneously (cross-layer optimization) can yield better performance than isolated compression techniques.
- (3) **Edge-cloud Collaborative Inference**- Split inference and collaborative AI, where the first few layers run on the device and the rest on the cloud, are promising approaches. Adaptive offloading policies can balance latency, energy, and bandwidth. For instance, real-time health monitoring systems can perform anomaly detection locally and detailed diagnosis in the cloud.
- (4) **Learning from Compressed Data**- In many IoT applications, input data (e.g., images, ECG, LIDAR) is compressed for transmission. Developing AI models that can operate directly on compressed representations could bypass decompression overheads and save resources. This includes learning from JPEG, video bitstreams, or compressed binary sensor logs.
- (5) **On-device Training and Personalization**- Personalized models can adapt to user behavior and device-specific patterns. However, on-device training is energy-intensive and requires careful memory management.

5 Conclusion: Toward Ubiquitous, Responsible AI

Ubiquity in AI refers not merely to widespread adoption but to seamless, context-aware integration into daily human activities. This entails designing AI systems that are interoperable across platforms, adaptive to diverse environments, and inclusive of varied user demographics. Achieving this requires robust infrastructure, scalable architectures, and edge intelligence that can operate efficiently in resource-constrained settings. Responsibility in AI encompasses transparency, accountability, fairness, and safety. As AI systems increasingly impact critical decision-making processes, it becomes imperative to incorporate mechanisms that ensure explainability and auditability. Approaches such as interpretable machine learning, differential privacy, and federated learning play a vital role in protecting user data and upholding ethical standards. Human-centricity places people, not technology at the core of AI design. It advocates for participatory design, where stakeholders are actively involved in shaping AI solutions that reflect their needs and aspirations. The pursuit of low-latency, low-power, and low-memory LLMs for edge deployment is not merely an engineering challenge, it is a foundational step toward making AI ubiquitous, accessible, and responsible. By enabling intelligent systems that operate locally, respect user privacy, and respond in real time, we can unlock a new generation of applications that are both powerful and human-centric. It further demands a shift in evaluation paradigms—from accuracy-centric to efficiency-aware metrics. As we continue to advance the capabilities of large language models at the edge, it is crucial to remain cognizant of their broader societal impact, ensuring these technologies are harnessed to empower individuals and communities in ways that are both equitable and sustainable.

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